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Generic or Quantum-Specific? Why the Control Matters: Difficulty-Matched Probes of LLM Quantum-Code Failures

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Abstract

When a large language model (LLM) fails to generate correct quantum code, is the failure quantum-specific—rooted in phase, basis, unitarity, or entanglement reasoning—or a generic code-generation weakness that merely surfaces on quantum tasks? We probe this with a paired-prompt design: every quantum-coding task is matched to a classical-coding task that exercises the same general programming skills without the quantum-specific content. Evaluating nine models from four providers on twelve paired prompts, graded by simulator equivalence, we record seventy-three quantum-half failures. Our central finding is methodological: the choice of control determines the answer. Against a structural control matched only on output shape, every observed quantum failure looks quantum-specific; but that control passes 99.5% of all cells, so the figure reflects its easiness, not quantum-specific cognition. We therefore construct a difficulty-matched control for the one prompt that admits a clean classical analogue—the three-qubit quantum Fourier transform, matched to the iterative Cooley–Tukey fast Fourier transform—under which apparent specificity falls to eight of sixteen (50%) and splits by model: capable models fail the quantum task while passing the classical ordering task, whereas weaker models fail both. For forty-four of sixty-three failures the quantum target is trivial to write classically, so no difficulty-matched control can exist and specificity is not measurable; we decline to claim those failures are quantum-specific. We release the dataset, cached responses, codings, and the full analysis pipeline.

Keywords: Quantum computing, Large language models, Code generation, Benchmarking, Failure-mode analysis, Qiskit

1 Introduction

Large language models (LLMs) now author a substantial fraction of newly written classical code, and the same models are being applied to *quantum* code (Qiskit, OpenQASM). A growing benchmark literature (Qiskit HumanEval [1], QuanBench [2], QuanBench+ [3], QCircuitBench [4], QSpark [5]) measures end-to-end task success on quantum prompts, and a parallel agentic literature (Quasar [6], Agent-Q [7], LLM4PQC [8], test-time learning [9]) attempts to fix the failures these benchmarks surface. Both lines of work implicitly assume a property they do not test: that the observed failures are *quantum-specific* — driven by phase, basis, unitarity, or entanglement-reasoning deficits — rather than *generic* code-LLM weaknesses (API recall, indexing, control flow) that happen to surface on quantum tasks because quantum APIs are underrepresented in pretraining corpora. The recent field review of generative AI for quantum circuits [10] explicitly catalogs this open question.

The distinction is load-bearing for three downstream decisions:

1. **Targeted training-data interventions.** If failures are mostly generic, "include more Qiskit code in pretraining" is the right intervention. If they are quantum-specific, the gap is conceptual: more Qiskit code in pretraining will not close it.
2. **Verifier-in-the-loop tooling.** Agentic systems with simulator feedback should target the failure modes that actually occur. Whether feedback recovers a failure depends sharply on whether the underlying cognitive shape is gate-ordering, recipe-verification, or recipe-commitment.
3. **Cognitive interpretation of LLM behavior.** Quantum code is an unusually distant out-of-distribution task for current models — the gates, semantics, and indexing conventions are far from anything in general code pretraining. Quantum-characteristic failures, if observed, constitute evidence that LLMs reason rather than pattern-match on tasks where in-distribution heuristics are more obviously inapplicable.

Contribution. We introduce a *paired-prompt* benchmark in which every quantum-coding task is matched with a classical-coding control that exercises the same general programming skills (function-signature handling, length-2^n structure construction, $1/\sqrt{2^n}$ normalization arithmetic, library API recall, stated-convention adherence) without requiring quantum-specific knowledge. A quantum failure for which the matched classical control *passed* at the same (prompt, seed, temperature) cell is evidence of a quantum-specific deficit — *but only to the extent the control is difficulty-matched*. We show that this caveat is load-bearing: a structural control inflates apparent specificity to ∼100%, while a difficulty-matched control (the classical FFT, for our QFT-3 case study) cuts it to 50% and reveals that the specificity is concentrated in the more capable models. Treating control design as a first-class variable is the paper's core methodological contribution.

We evaluate 9 models across 4 providers (Anthropic Claude Opus 4.7 / Sonnet 4.6 / Haiku 4.5; OpenAI GPT-5.5 / 5.4-Mini; Google Gemini 2.5 Flash / Flash-Lite; Qwen2.5-Coder-32B and Llama-3.3-70B routed via HuggingFace Inference) on 12 paired prompts spanning 5 difficulty tiers. We hand-code every quantum failure into a multi-coder taxonomy. Finally, we run a simulator-feedback agent on the failure cells

to test whether different failure subtypes are differently recoverable. The dataset, all cached LLM responses, the failure-mode codings, and the analysis pipeline are released for reproducibility.

2 Related Work

Quantum-code benchmarks. Qiskit HumanEval [1] introduced the canonical Qiskit-targeted benchmark in the spirit of HumanEval [11], with 151 tasks. QuanBench [2] and QuanBench+ [3] extend to multi-framework evaluation. QCircuitBench [4] provides a large-scale circuit dataset. QSpark [5] focuses on output reliability. None of these benchmarks construct matched classical controls.

Agentic systems with simulator feedback. Quasar [6], Agent-Q [7], LLM4PQC [8], and test-time-learning approaches [9] treat circuit synthesis as a closed-loop optimization with the simulator as verifier. Our work intentionally uses such an agent only as a closing demonstration; the contribution is the controlled measurement of failure specificity.

Code-LLM failure-mode analysis. Where-do-LLMs-still-struggle [12] catalogs four failure modes (wrong-problem-mapping, flawed-algorithm, edge-case mishandling, formatting) for classical code-generation tasks. Failure-aware enhancements [13] pair empirical failure analysis with mitigations. Prompt-specificity [14] probes sensitivity to prompt detail. None of these target quantum code.

Field reviews. Generative AI for quantum circuits [10] surveys the field but does not propose a controlled measurement of failure specificity; this paper directly addresses one of its open questions.

3 Methods

3.1 Paired-prompt design

A *paired prompt* consists of a quantum task Q (Qiskit code completion producing a target unitary or statevector) and a classical task C (NumPy code completion producing the matched length-2^n vector or matrix). The two halves share function-signature shape, output structure size, normalization arithmetic, and library-API-recall demand. The classical half explicitly does not require quantum-specific knowledge (phase, basis, unitarity, entanglement); the quantum half does.

3.2 Transferable vs. quantum-native difficulty

Whether a *difficulty-matched* classical control can be constructed at all depends on where the quantum task's difficulty comes from. We distinguish two regimes, and this distinction organizes the rest of the paper.

Transferable-difficulty tasks. The quantum task is hard for a reason that survives translation into a classical algorithm on real-valued data. The 3-qubit QFT is the cleanest case: its difficulty is the staged decomposition into ordered, non-commuting operations, and that difficulty is shared structure-for-structure by the classical Cooley–Tukey FFT. Grover's iteration is similar — oracle-then-diffusion is a pair of reflections

that act identically on a real amplitude vector. For such tasks a difficulty-matched control exists and the quantum-specificity ratio is directly measurable.

Quantum-native-difficulty tasks. The target object is trivial to produce classically — a one-line matrix or vector — and *all* of the difficulty lies in expressing it through the quantum gate set. The phase-flip oracle's target is `numpy.eye(8)` with one entry negated; the alternate Bell state is $[1, 0, 0, -1]/\sqrt{2}$; the Toffoli is a fixed permutation matrix. Writing these classically is trivial (our structural controls pass them 99.5% of the time), so no *difficulty-matched* classical control can exist — there is no hard classical task to match against. **This is a scoping statement, not a specificity result.** For these tasks our control-based test is simply *inapplicable*: we cannot rule out that the failures are generic recall or instruction-following errors. We are explicit that this cuts against us — our independent re-coder (Section 3.9) codes the entire alternate-Bell (Φ^-) family as *generic* wrong-gate errors, and that family is 10 of the 44. So we do *not* assert these failures are quantum-specific; we claim only that their difficulty has no classical-amplitude analog (a relative sign or phase is not observable on classical probabilities), which is a statement about the *task*, not a measured property of the failures.

This partition reframes the near-ceiling structural controls (Section 4.3): for transferable tasks they are uninformative and must be replaced by a matched control (we do so for QFT-3); for quantum-native tasks their triviality marks where a control-based test cannot run. Of our 63 genuine wrong-unitary failures (we separate these from 10 execution failures in Section 3.6), **44 (70%)** fall on these control-inadmissible quantum-native tasks (phase-flip oracle, `phase_flip_010`, Toffoli decomposition, Bell-Φ^-) and 19 on transferable tasks (QFT, Grover, parameterized rotation) — so the honest headline is that for most failures we *cannot* measure specificity, and the one task where we can (QFT-3) gives 50%.

3.3 Tier ladder

We organize prompts by difficulty:

- **Tier 1.** Uniform single-gate application (Hadamard fan-out; uniform-vector construction).
- **Tier 2.** Per-element parameterized rotation (RY with half-angle relation to amplitude; matched tensor-product vector).
- **Tier 3.** Two-qubit and multi-qubit entanglement (Bell, GHZ; matched explicit basis-vector targets).
- **Tier 4.** Multi-qubit phase reasoning (QFT decomposition; phase-flip oracle; matched matrix targets and explicit DFT).
- **Tier 5.** Fixed-iteration algorithm composition: the 2-qubit, single-iteration Grover instance. We flag honestly that this is the *deterministic special case* — with one marked item in four, a single iteration rotates exactly onto the marked state — so it tests recall of a canonical (likely in-distribution) circuit rather than iteration-count reasoning, the part of Grover that is most distinctively quantum. A multi-iteration Grover where the iteration count matters is future work.

3.4 Models

We evaluate nine models from four providers spanning frontier paid, frontier free-tier, and open-weights: Anthropic Claude Opus 4.7, Sonnet 4.6, Haiku 4.5; OpenAI GPT-5.5, GPT-5.4 Mini; Google Gemini 2.5 Flash, Flash-Lite; Qwen2.5-Coder-32B and Llama-3.3-70B via the HuggingFace Inference Router.

3.5 Decoding

For each (model, prompt-half) pair we sample at temperature $T = 0.0$ (one deterministic-where-possible run) and at $T = 0.7$ across three random seeds $\{0, 1, 2\}$. We disable Gemini 2.5's hidden *thinking* budget by setting `thinking_budget = 0`. We *intended* to set OpenAI GPT-5.x to `reasoning_effort = minimal` as well, but discovered (during the audit behind this version) that GPT-5.x rejects the `temperature`/`seed` parameters and our retry path silently dropped `reasoning_effort` along with them — so **GPT-5.x actually ran at the API default reasoning effort**, not minimal. We have fixed the client, disclose the issue here, and did not re-run the full panel. The practical consequence is that only *Gemini* was genuinely reasoning-suppressed; GPT-5.x got full default reasoning, so the decoding asymmetry is smaller than a naive reading suggests, and the thinking-budget ablation (Section 4.4) is meaningful only for Gemini — where all 6 runnable suppressed-budget failures persist at full budget. Anthropic Claude 4.7 deprecated the `temperature` parameter; we record the nominal (T, seed) label for those calls but the API uses Anthropic-default sampling.

3.6 Grading

Each model output is parsed for fenced Python code, executed in a fresh namespace, and graded against a hand-written test that uses `Operator.equiv` (circuit-unitary tasks), `Statevector.equiv` (state-preparation tasks), and `numpy.testing.assert_allclose` (classical-half tasks). Equivalence is checked *up to a global phase*, consistent with the physical equivalence class of unitaries; we verified this credits global-phase variants (e.g. $\text{diag}(1, \ldots, -1)$ vs. its negation) while still rejecting wrong-target-state flips and distinguishing relative-phase errors ($|\Phi^-\rangle$ vs. $|\Phi^+\rangle$).

We separate *cognitive* failures (grader class `test_failed`: a well-formed circuit whose unitary is wrong — 63 of our 73 quantum failures) from *execution* failures (syntax/runtime errors — 10 failures, several of them completions truncated by an initial HuggingFace token-budget bug, since fixed and re-run; the re-run cells remained failures, confirming they are genuine rather than truncation artifacts). Only the 63 cognitive failures enter the taxonomy and the transferable/quantum-native partition. The "primitive gates only" constraint is enforced by instruction and audited post-hoc, not by the grader. All outputs are cached on disk, keyed by $\text{SHA256}(\text{prompt}, \text{model}, T, \text{seed})$ for reproducibility.

3.7 Primary metric: quantum-specificity ratio

For each model, of all its *quantum* failures, what fraction had a matched classical control *pass* at the same (prompt, seed, T)? A high ratio is intended to indicate quantum-specific failure. This metric is only as good as the control: it is informative *only if the classical control is difficulty-matched*, so that passing it genuinely rules out a generic deficit. We pre-registered a one-sided binomial test (null rejected at $p < 0.01$ for ratio $\geq 30\%$); we report it for completeness but de-emphasize it, because (i) our structural controls turn out to be near-ceiling (Section 4.3), which makes the ratio an upper bound rather than a measurement, and (ii) the failures are clustered — spread across seven prompts but concentrated in the Tier-4/5 ladder, and repeated across seeds and models — so they are not the independent Bernoulli trials the test assumes. The difficulty-matched control below is our remedy.

3.8 Difficulty-matched ("stress") control

A structural control shares the quantum task's output *shape*; a difficulty-matched control shares the *cognitive operation* whose failure we are trying to attribute. For our flagship QFT-3 case study, the dominant quantum failures are *ordering* errors (Section 4.5): the model knows the gate set but emits the H and controlled-phase gates in the wrong order. We therefore add a classical control that demands the same skill — producing the correct ordered sequence of staged, non-commuting transforms — with no quantum semantics: the iterative radix-2 Cooley–Tukey FFT. The task returns the intermediate state after each butterfly stage, so a correct answer requires the staged decomposition itself and *cannot* be recovered from the direct $O(N^2)$ DFT formula (we verify that a direct-formula solution fails the grader). If a model passes this control but fails the QFT at the same cell, the QFT failure is isolated to quantum-specific reasoning rather than to generic difficulty at ordered composition of non-commuting operations. We construct this matched control for the QFT-3 case study; extending it to every prompt is future work (Section 6).

3.9 Failure-mode taxonomy

Each quantum-half failure is first coded by the author (the "primary" coding): a heuristic auto-classifier (`src/handcode.py`) provides a starting pass and every cell is then read and confirmed or corrected by hand. Categories were assigned observation-first: a label is created when a representative example is identified, and subsequent matching cells are added to the existing label rather than spawning new ones.

To test whether the codebook is reproducible, we obtain independent re-codings from two OpenAI models (GPT-5.5 and GPT-5.4 Mini), each given only the codebook and the generated code, blind to the primary labels and to each other. (We also attempted cross-provider raters via Google Gemini and several OpenRouter models, but free-tier rate limits left their coverage too incomplete to use.) These raters are LLMs, not humans; we report *automated* inter-rater reliability and leave human re-coding to future work. Agreement is *capability-dependent*: the capable GPT-5.5 reaches Cohen's $\kappa = 0.68$ with the author's coding (substantial, just under the $\kappa \geq 0.7$ pre-registered in our protocol), whereas the smaller GPT-5.4 Mini agrees less ($\kappa = 0.39$),

which pulls the three-rater Fleiss' κ down to 0.45 (95% bootstrap CI [0.36, 0.53]). We read this as evidence that fine-grained quantum-failure coding is itself a capability-sensitive task, and that the *subtype* codebook is substantially reproducible by a capable independent rater.

We stress one limitation of this κ: it measures agreement on the fine subtypes (F2a, F4c, ...), *conditional* on a failure being a cognitive one. It does *not* validate the binary quantum-specific-vs-generic axis the framework leans on, because the author's primary coding places all 63 cognitive failures in the quantum-specific branch and none in the generic branch — so at the branch level Cohen's κ is undefined (one rater is constant). On that axis the independent raters actively disagree: GPT-5.5 codes the 10 alternate-Bell (Φ^-) failures as *generic* `wrong-gate` errors, and GPT-5.4 Mini crosses 10 cells to generic. The quantum-specific-vs-generic call therefore has no measured inter-rater reliability, and we do not claim it does. Per-rater assignments are released at `results/coders.json`.

Beyond the QFT (F2) and phase-oracle (F4) families detailed below, the remaining Tier-4/5 prompts contribute their own families — decomposition (F5), Grover oracle/diffusion (F6), and missing-sign-flip (F7) — which we name but do not case-study in depth.

Quantum-specific subtypes.

F2a — Gate-cluster ordering (QFT). All Hadamard gates emitted at the start; all controlled-phase gates clustered after. Correct gates and angles, wrong interleaving.
F2b — Reversed iteration direction (QFT). Iterates qubits $0 \to n-1$ instead of $n-1 \to 0$. Each H precedes its targeting controlled-phases, but the recursion direction is flipped.
F2c — Spurious extra phases (QFT). Emits additional controlled-phase gates beyond what the algorithm requires.
F3a — Asymmetric edge-case guard (RY parameterization). Skips a parameterized rotation when $p \in \{0, 1\}$. Correct at $p = 0$, incorrect at $p = 1$.
F4a — X-conjugation onto wrong basis state (phase oracle). Wraps a correct phase-kickback core in X gates; conjugation moves the phase flip onto a different basis state.
F4b — No-op primitive substitution (phase oracle). Inserts a single- qubit Z acting on a $|0\rangle$ qubit. Composition is the identity.
F4c — Identity-circuit emission (phase oracle). Emits a structurally plausible recipe (CCX + X gates) where every gate is its own inverse and the composition exactly cancels.
F4d — Incoherent thinking (phase oracle). Long verbose comments, contradictory steps, or multiple `QuantumCircuit()` restarts mid- response without committing to a recipe.

Generic subtype.

F3b — LSB/MSB convention error (classical half). Uses the bit- ordering convention opposite to the prompt's stated convention. We defined this category a priori as our example of a generic failure; it was observed *zero* times in the data, which is itself

telling — the author's coding found no clearly generic cognitive failure, the degeneracy discussed above.

3.10 Statistical methods

All statistics are recomputed from the cached data by `src/stats.py`, `src/modeling.py`, and `src/report.py` (each function checked against hand-computed values in `tests/`). We report Wilson score intervals for proportions; for the 100% models we add the rule-of-three upper bound (0/96 implies a true failure rate below 3.1%) rather than claiming a literal zero. For the matched-control specificity we report both a per-cell Wilson interval and a *cluster bootstrap* that resamples whole models, since the failure cells are not independent. We test the difficulty (tier) effect with a logistic regression using cluster-robust standard errors by prompt, complemented by a cell-level Spearman trend; model heterogeneity with a χ^2 test of model $\times$ outcome; the agent uplift with the pre-registered exact McNemar test (the correct paired test); and taxonomy reliability with Cohen's and Fleiss' κ plus a bootstrap κ interval. We deliberately avoid the per-cell binomial test the protocol pre-registered, because the cells are clustered and non-independent.

3.11 Use of generative AI in the research process

In line with Springer Nature editorial policy, we document the use of generative AI in producing this work. The research was conceived, designed, and directed by the corresponding author (M.G.), who formulated the central question, designed the paired-prompt methodology and the difficulty-matched control, selected and interpreted the analyses, and is fully accountable for the results. A large language model (Anthropic Claude) was used as an assistant under the author's direction: it supported implementation of the evaluation harness and analysis scripts, performed routine statistical computation, and assisted in drafting and copy-editing the manuscript. All code was specified, reviewed, and tested by the author; every reported number was regenerated from the cached data and checked by the author; and all scientific claims and their interpretation are the author's own. No large language model satisfies the journal's authorship criteria, and none is listed as an author. (The LLMs named elsewhere in this paper are the *objects of study*, not tools used to write it.)

4 Results

4.1 Headline pass rates

Table 1 reports per-model pass counts and rates over the full paid panel, and Figure 1 shows the same results split by the quantum and classical halves.

4.2 Per-tier behavior

Two inferential checks support what Figure 2 shows (Section 3.10). Models differ in quantum-half failure rate far beyond chance ($\chi^2(8) = 77$, $p = 2 \times 10^{-13}$; this treats cells as independent, so the p is optimistic given prompt clustering, but the effect is

Table 1 Per-model pass counts and rates across 12 paired prompts × 2 halves × up to 4 calls (1 greedy + 3 stochastic seeds at $T = 0.7$), sorted by pass-rate. "Err" counts cells where the API errored (Gemini free-tier 429s on stochastic calls); these are excluded from the pass-rate denominator. Two models (Claude Sonnet 4.6, GPT-5.5) reach 100%; by the rule of three, 0/96 implies a true failure rate below 3.1% at 95% confidence, so we read "100%" as that upper bound rather than a literal zero. CIs are Wilson score.

Model	Pass	Fail	Err	Pass-rate	95% CI
Claude Sonnet 4.6	**96**	**0**	0	**100.0%**	[96.2, 100.0]
GPT-5.5	**96**	**0**	0	**100.0%**	[96.2, 100.0]
Claude Opus 4.7	94	2	0	97.9%	[92.7, 99.4]
Gemini 2.5 Flash	45	2	49	95.7%	[85.8, 98.8]
GPT-5.4 Mini	87	9	0	90.6%	[83.1, 95.0]
Gemini 2.5 Flash-Lite	42	5	49	89.4%	[77.4, 95.4]
Claude Haiku 4.5	83	13	0	86.5%	[78.2, 91.9]
Qwen2.5-Coder-32B	76	20	0	79.2%	[70.0, 86.1]
Llama-3.3-70B	71	24	1	74.7%	[65.2, 82.4]

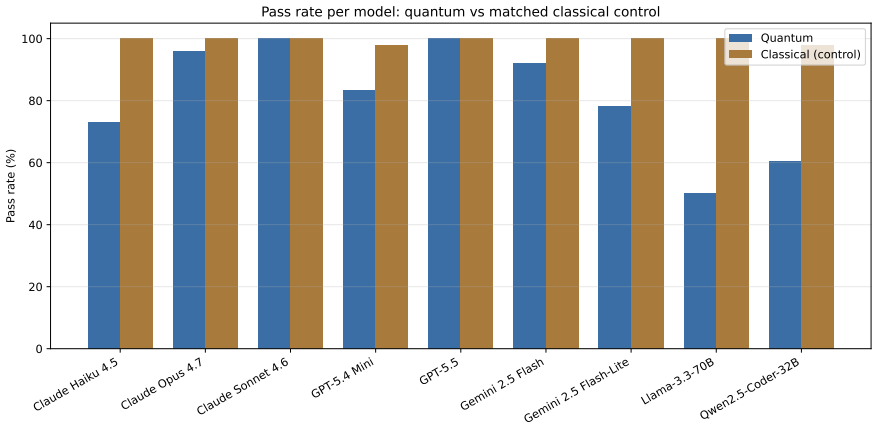


Fig. 1 Pass rate per model, split by half. The matched classical control passes at near-100% on every model; quantum-half deficits are visible only for the lower-tier models.

large enough that the conclusion is not in doubt). The difficulty trend is real at the cell level (Spearman $\rho = 0.26$, $p = 3 \times 10^{-7}$: higher tiers fail more) but, once we account for prompt clustering with a logistic regression using cluster-robust standard errors, the per-tier odds ratio of 1.87 is only suggestive (95% CI $[0.90, 3.89]$, $p = 0.09$). With twelve prompts the tier effect is not yet prompt-level significant; the strong, robust result is that *models* differ, not that we can pin the difficulty curve.

4.3 Specificity under a structural control (an upper bound)

For each model, of all its quantum-half failures, we ask: did the *structural* classical control — matched on output shape — *pass* at the same (prompt, seed, temperature)?

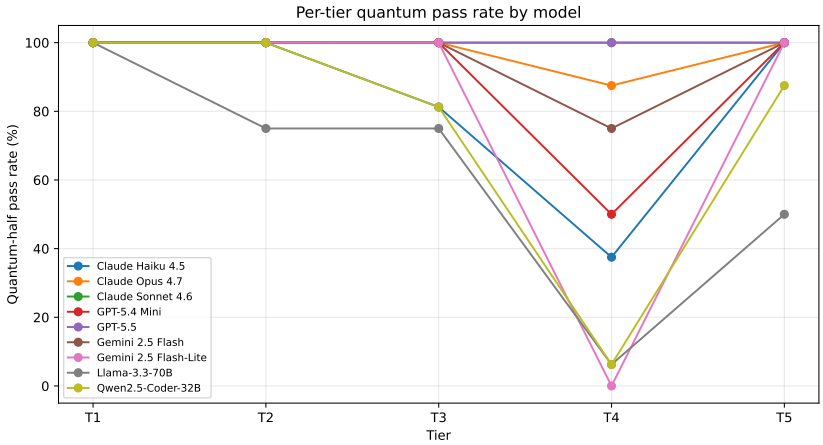


Fig. 2 Per-tier quantum-half pass rate by model. Tiers 1–3 (single-gate, parameterized rotation, two-qubit entanglement) are at ceiling for almost every model. Tier 4 (multi-qubit phase reasoning: QFT-3 and phase-flip oracle) is the principal source of failures across all but three frontier paid models.

Table 2 Specificity under the *structural* control, computed over *observed* matched pairs (a classical cell lost to a Gemini 429 is missing data, not a classical failure, and is excluded). The structural control passed at **every one** of the 71 quantum failures where it was observed: structural specificity is 71/71 = 100%. We stress this is **not** evidence of quantum-specificity — it is evidence that the structural control essentially never fails (it passes 99.5% of cells overall), so it cannot discriminate. The difficulty-matched control (Table 3) is the informative one. Full per-cell records at `results/summary.csv`.

Model	Q-failures (obs.)	Classical PASS	excl. (ctrl ERR)	Specificity
Claude Opus 4.7	2	2	0	100.0%
Claude Haiku 4.5	13	13	0	100.0%
GPT-5.4 Mini	8	8	0	100.0%
Gemini 2.5 Flash	1	1	1	100.0%
Gemini 2.5 Flash-Lite	5	5	0	100.0%
Llama-3.3-70B	23	23	1	100.0%
Qwen2.5-Coder-32B	19	19	0	100.0%
Total	**71**	**71**	**2**	**100.0%**

The pre-registered one-sided binomial test (against $H_0 : p_{\text{specific}} = 0.30$) is nominally passed by a wide margin, but we deliberately do *not* headline a p-value here: with a control that passes 99.5% of the time and 73 clustered, non-independent failures, the test is rejecting a straw-man null and its small p conveys little. Read literally, Table 2 says "a quantum failure *never* co-occurs with a structural-control failure" — which follows from the control being easy, not from the failure being quantum-specific. To make the question answerable we need a control that can fail on generic grounds; that is the next subsection.

On the Gemini missing-data exclusion.

Gemini free-tier HTTP 429s (daily quota) cost 48 cells on the quantum half and 50 on the classical half across Flash and Flash-Lite — including a few greedy ($T = 0$) cells, not only stochastic ones. We exclude these from the pass-rate denominators rather than impute them. The loss is *not* perfectly symmetric across the two halves (48 vs. 50), so for the matched-pair specificity we restrict to cells where *both* halves were observed; the Gemini specificity is therefore computed on that smaller observed-pair set, and we flag the two Gemini rows in Table 2 as the least data-complete.

4.4 Specificity under a difficulty-matched control

We re-ask the specificity question for the QFT-3 case study using the difficulty-matched FFT control of Section 3.8, run on the full panel at the same (prompt, seed, temperature) cells.

Table 3 QFT-3 under the difficulty-matched control. "Quantum" and "Structural" are pass rates on the QFT and on the transcription control; "Matched" is the FFT butterfly-ordering control. The structural control is at ceiling for every model; the matched control is at ceiling for the frontier models (so it is fair, not impossibly hard) but *separates* the weaker ones. A QFT failure is isolated as quantum-specific only when the model passes the matched control at the same cell.

Model	Quantum	Structural ctrl	Matched ctrl
Claude Opus 4.7	4/4	4/4	4/4
Claude Sonnet 4.6	4/4	4/4	4/4
GPT-5.5	4/4	4/4	4/4
Gemini 2.5 Flash	4/4	4/4	4/4
GPT-5.4 Mini	2/4	4/4	4/4
Claude Haiku 4.5	1/4	4/4	4/4
Qwen2.5-Coder-32B	0/4	4/4	3/4
Gemini 2.5 Flash-Lite	0/4	4/4	0/4
Llama-3.3-70B	1/4	4/4	1/4

Figure 3 plots the QFT-3 pass rates under the quantum task and the two controls. Aggregating over the QFT-3 failures, only **8 of 16 (50%, 95% Wilson CI [28, 72])** occur at a cell where the model *passed* the matched control — half the apparent specificity the structural control reported. Because these 16 cells cluster within only 5 models, the per-cell Wilson interval understates the uncertainty; a cluster bootstrap that resamples whole models gives the honest and much wider interval [12, 93]%. The 50% point estimate should therefore be read as "about half, with large model-level uncertainty," not a precise fraction. The split is model-dependent: Claude Haiku 4.5 (3/3) and GPT-5.4 Mini (2/2) fail the QFT while passing the classical ordering task, so their QFT failures are genuinely quantum-specific, and Qwen2.5-Coder-32B is mostly so (3/4). By contrast, Gemini 2.5 Flash-Lite (0/4) and Llama-3.3-70B (0/3) fail the matched control too — they cannot reliably order the non-commuting *classical* butterfly stages either, so their "quantum" failures on QFT-3 are better read as generic difficulty with ordered composition than as a quantum-specific deficit. This

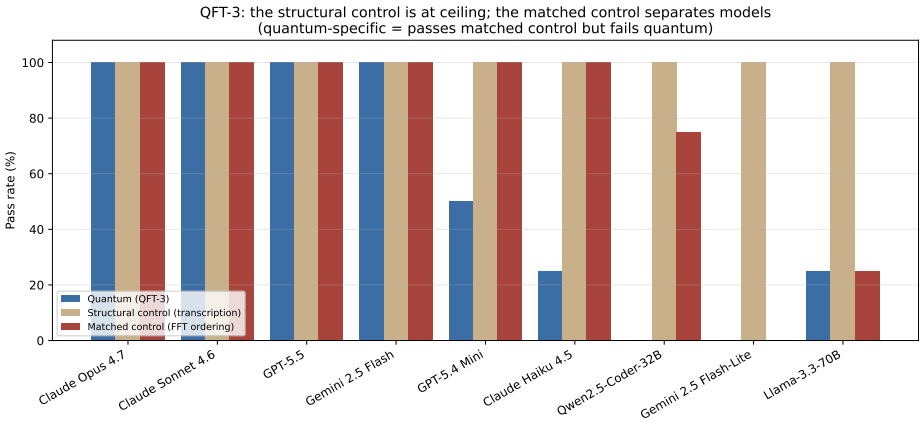


Fig. 3 QFT-3 pass rates under the quantum task, the structural control, and the difficulty-matched control. The structural control is a flat ceiling; the matched control separates models. The gap between the matched-control bar and the quantum bar is the genuinely quantum-specific deficit.

is the paper's central lesson: *the same failures read as ∼100% or 50% quantum-specific depending only on how hard the control is*, and a trivial control systematically over-attributes failures to quantum-specific cognition.

Robustness: the failures are not prompt-wording artifacts.

A fair worry is that a failure reflects one unlucky phrasing rather than a stable deficit. We re-ran every model that failed the original QFT-3 prompt on a semantically identical but reworded prompt (same signature and grading test; `src/ablation.py`). **13 of 16 (81%)** of the original failures persist under the paraphrase — including all of Qwen2.5-Coder-32B's and Gemini 2.5 Flash-Lite's — so the QFT-3 failures are largely wording-invariant. The three that flipped (one Claude Haiku 4.5, two Llama-3.3-70B) indicate some residual prompt sensitivity, which we report rather than hide.

Robustness: reasoning-budget suppression.

Only Gemini was genuinely reasoning-suppressed (GPT-5.x ran at default effort due to the client bug in Section 3.5, so its budget never varied and we exclude it from this ablation). We re-ran Gemini's suppressed-budget failures at full thinking budget (`src/ablation_thinking.py`): all **6 of 6** runnable cells (Flash-Lite 5/5, Flash 1/1; one Flash cell was rate-limited) *still fail*. So even the one genuinely budget-suppressed family is not a budget artifact. Crucially, GPT-5.4 Mini's two QFT-3 failures — part of the flagship matched-control result — were produced at full default reasoning, so that result does not depend on any suppression.

4.5 Case study: QFT-3 (Tier 4)

The 3-qubit QFT prompt produces 16 of the 63 wrong-unitary failures across 5 distinct models (all 16 are genuine wrong-unitary, none execution errors). Per-cell results:

QFT-3 failures partition into three cognitively-distinct subtypes:

Table 4 QFT-3 quantum half, per-cell pass/fail. Greedy at $T = 0.0$, then 3 seeds at $T = 0.7$. All cells now data-complete.

Model	greedy	T=0.7, s=0	T=0.7, s=1	T=0.7, s=2	rate
Claude Opus 4.7	✓	✓	✓	✓	4/4
Claude Sonnet 4.6	✓	✓	✓	✓	4/4
GPT-5.5	✓	✓	✓	✓	4/4
Gemini 2.5 Flash	✓	✓	✓	✓	4/4
GPT-5.4 Mini	×	✓	×	✓	2/4
Claude Haiku 4.5	✓	×	×	×	1/4
Llama-3.3-70B	✓	×	×	×	1/4
Gemini 2.5 Flash-Lite	×	×	×	×	0/4
Qwen2.5-Coder-32B	×	×	×	×	0/4

Gate-clustering (F2a): All Hadamard gates emitted at the start of the circuit, all controlled-phase gates emitted afterwards. Correct gates, correct angles, correct swap; wrong interleaving. The standard QFT decomposition requires $H(i)$ *before* the controlled-phase gates targeting qubit i; clustering breaks the unitary because gates sharing qubits do not commute.
Reversed iteration direction (F2b): Iterates qubits $0 \to n{-}1$ with controlled-phases targeting qubit 0 first, instead of $n{-}1 \to 0$. Each Hadamard is correctly placed before its targeting phases, but the recursion direction is reversed.
Spurious extra phase gates (F2c): Correctly interleaves H and controlled-phase gates for the first portion, then emits additional controlled-phase gates not present in the algorithm.

4.6 Case study: phase-flip oracle (Tier 4)

The phase-flip oracle prompt (implement $\mathrm{diag}(1, 1, \ldots, 1, -1)$ on 3 qubits via primitive gates) produces 11 failures across 5 distinct models, of which 7 are wrong-unitary (the cognitive subtypes below) and 4 are execution errors. Three named subtypes appear:

X-conjugation onto wrong state (F4a): Wraps a correct phase-kickback core in $X^{\otimes 3}$, which conjugates the target state and ends up flipping the bit-complement basis state instead of $|111\rangle$. Cognitive shape: knows there is a "convert-act-undo" trick but mistracks which state is being flipped under conjugation.
No-op primitive substitution (F4b): Builds the right outer scaffolding (X-bracketing plus a CCX-based controlled-phase) but substitutes a single-qubit Z acting on a $|0\rangle$ qubit, where $Z|0\rangle = +|0\rangle$. The composed unitary is the identity. Cognitive shape: vague memory of the H-CCX-H trick replaced with a primitive that does not actually implement it.
Identity-circuit emission (F4c): Emits a structurally plausible recipe (multiple controlled-flip gates, X bracketing) where every gate is its own inverse and the composition exactly cancels. Cognitive shape: produces the visual form of a phase-flip recipe without verifying that the composition introduces any phase.

Figure 4 breaks each model's quantum-half failures into the named subtypes defined above.

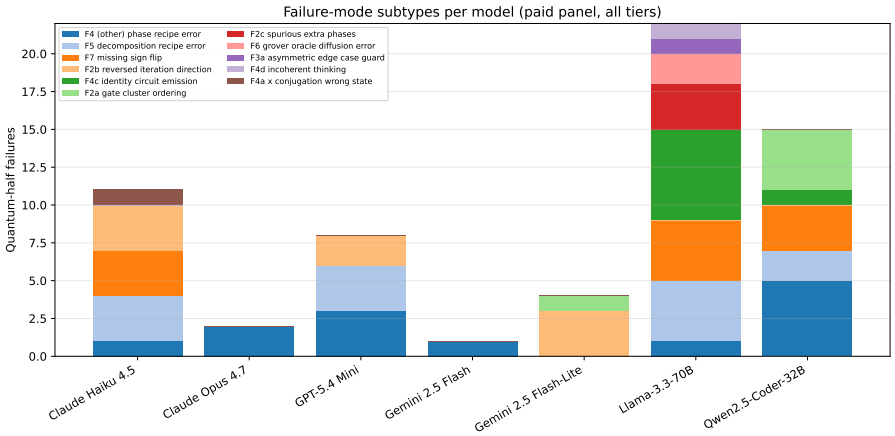


Fig. 4 Failure-mode subtypes per model (paid panel, all tiers). Each bar breaks the model's quantum-half failures into the named taxonomy categories defined in Section 3.9. Two models (Claude Sonnet 4.6, GPT-5.5) have no bar because they have no failures.

5 Discussion

We organize the discussion around four threads: (1) the cognitive shape underlying the observed failures, (2) the model hierarchy and what appears to predict pass-rate, (3) the value of multi-seed evaluation, and (4) the agent uplift result and the F2-vs-F4 recovery asymmetry.

5.1 Cognitive shape: visual templates without semantic verification

The eight quantum-specific subtypes we observe (F2a/b/c on QFT-3, F4a/b/c/d on the phase-flip oracle, F3a on parameterized rotation) converge on a single broader cognitive pattern: *LLMs reproduce the visual surface of a quantum-code recipe without verifying that the resulting unitary computes the claimed transformation.*

Concrete instances of this pattern across our subtypes:

- **F2a (gate-clustering)**: emits the right *list* of gates with the right angles, but the *interleaving* is wrong. The model has memorized "QFT looks like H...H, CP...CP" as a syntactic shape rather than the semantic constraint that $H(i)$ must precede the controlled-phases targeting qubit i.
- **F2c (spurious extra phases)**: emits a syntactically plausible prefix that locally interleaves correctly, then continues to emit additional controlled-phase rotations that the algorithm does not require. The model has learned that QFT has "many CP gates" but not when to stop.
- **F4b (no-op primitive substitution) and F4c (identity-circuit emission)**: emits a circuit that has the visual texture of a phase-flip recipe (X gates bracketing a multi-controlled gate) but where the composition is the identity. The model has not checked that any non-trivial phase is actually introduced.

- **F4d (incoherent thinking)**: emits a stream of recipe-shaped comments and reset attempts without committing to any single recipe. The model recognizes it "ought" to produce a phase-flip recipe but cannot commit to one.

This shared pattern complements existing classical code-LLM failure-mode analyses [12–14] that target algorithmic correctness rather than physical-system correctness. The strength of the "surface-form without verification" reading depends on the control, exactly as Section 4.4 shows: where a model passes the *difficulty-matched* control but fails the quantum task (Claude Haiku 4.5, GPT-5.4 Mini on QFT-3), the deficit is cleanly quantum-specific and the reading holds; where the model fails the matched control too (Gemini 2.5 Flash-Lite, Llama-3.3-70B), the same surface-form failure is better explained by a generic weakness at ordered composition. The pattern is therefore most defensible for the more capable models.

5.2 Model hierarchy and what predicts pass-rate

Two models achieve perfect 100% pass rates across the benchmark: Claude Sonnet 4.6 and GPT-5.5 — both current-generation flagship models from major frontier labs. Claude Opus 4.7 is essentially perfect (97.9%; 2 of 96 cells failed). Llama-3.3-70B (74.7%) and Qwen2.5-Coder-32B (79.2%) occupy the bottom of the table, separated from the frontier models by 20+ percentage points.

Two patterns are worth flagging:

1. **Code-specialization is not sufficient (with a caveat).** Qwen2.5-Coder-32B is fine-tuned on code; it nonetheless underperforms the generalist Claude Sonnet 4.6 by more than 20 percentage points. This is *consistent* with quantum code requiring skills beyond general code competence — but it is confounded by scale: Qwen is a 32B open-weights model and Sonnet is a much larger frontier model, so we cannot separate "specialization is not enough" from "Qwen is simply smaller." We report the gap and flag the confound rather than over-claim a clean specialization effect.
2. **Within-family scaling holds, but not perfectly.** Claude Sonnet 4.6 outperforms Claude Haiku 4.5 by 13 percentage points (100.0 vs. 86.5), and Claude Opus 4.7 sits between them at 97.9% (2 failures out of 96 cells, versus Sonnet's 0). The non-monotonicity is small in absolute terms and may not survive a larger benchmark, but two artifacts of our methodology may also contribute: (a) Anthropic deprecated the `temperature` parameter on Claude 4.x at the time of the run, so the nominal $T = 0.7$ stochastic seeds for Opus and Sonnet draw from Anthropic-default sampling rather than a user-controlled distribution; (b) both Opus failure cells are on the new Tier-4 prompts (toffoli_decomp), where Opus's reasoning may interact with the prompt's library-shortcut prohibition differently than Sonnet's. Within OpenAI, GPT-5.5 outperforms GPT-5.4 Mini by 9 percentage points (100.0 vs. 90.6). Within Google, Gemini 2.5 Flash outperforms Flash-Lite by 6 percentage points on the cells where stochastic data is available.

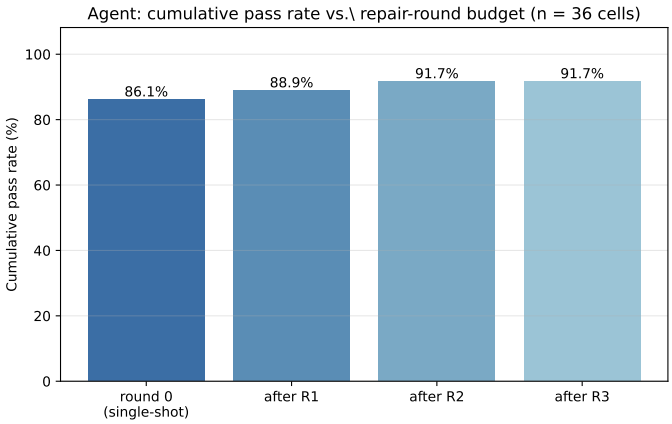


Fig. 5 Agent uplift over rounds. Cumulative pass rate across the 36 cells (12 prompts × Claude Opus / Sonnet / Haiku, quantum half) at each repair-round budget. Round 0 = single-shot baseline; subsequent rounds add structured grader feedback.

5.3 Closing demonstration: simulator-feedback agent

As a closing demonstration — explicitly not a methodological contribution — we wrap the LLM in a minimal feedback loop. After the round-0 single-shot generation (the standard runner output), if the test fails, we feed the grader's error message back to the model in the same conversation thread and request a corrected version. We allow up to 3 repair rounds (4 total attempts). The feedback contains only the structured grader output (error class and assertion message), *not* the canonical solution or hints about which gate the model should change.

The implementation (`src/agent.py`) is OpenAI-compatible and ready to run against any of our model panels. The intended cost-neutral target is the free-tier OpenRouter panel; daily request quotas on the free tier (50 requests / account / day; 250 across the five keys we rotate across) are tight relative to a full agent sweep over our 12 prompts, so the headline agent run is reported on the paid panel, where rate limits do not bind.

Headline result: 12 prompts × 3 Claude tiers = 36 cells.

Figure 5 plots the cumulative pass rate at each repair-round budget. Round 0 (single-shot) passes: 31/36 (86.1%). After up to 3 repair rounds: 33/36 (91.7%). Two cells recovered by feedback (failed round 0, passed eventually); three cells were unrecoverable within the 3-round budget. The uplift is not statistically significant at this scale: the pre-registered exact McNemar test on the discordant pairs (2 recovered, 0 regressed) gives $p = 0.50$. The minimum detectable effect is large here: with zero regressions, even a perfect recovery of both failing cells could not reach $p < 0.05$ at this n. We therefore read the agent result as strictly directional — consistent with, but not confirmation of, the F2-vs-F4 recovery asymmetry below.

Prompt	Model	R0	R1	R2	R3	Outcome
QFT-3 (T4)	Claude Haiku 4.5	×	✓	—	—	**recovered (R1)**
phase-oracle (T4)	Claude Haiku 4.5	×	×	×	×	persistent
phase_flip_010 (T4)	Claude Opus 4.7	×	×	✓	—	**recovered (R2)**
phase_flip_010 (T4)	Claude Haiku 4.5	×	×	×	×	persistent
toffoli_decomp (T4)	Claude Haiku 4.5	×	×	×	×	persistent

The F2-vs-F4 recovery asymmetry holds at full panel scale.

The single recovered F2-family failure (QFT-3, Haiku) was repaired after one repair round given a structured grader-error message. Among F4-family failures (phase-oracle and the alternate-state phase_flip_010 prompts), 1 of 3 recovered (Opus on phase_flip_010, at round 2) and 2 of 3 persisted across the full 3-round repair budget (both Haiku). The toffoli_decomp failure (Haiku) similarly persisted: this prompt requires a textbook 6-CNOT decomposition with delicate $T/T^\dagger$ placement; a round-by-round error message is unlikely to help a model that does not already remember the recipe.

These observations support the cognitive interpretation laid out in Section 3.9: F2 failures (gate-ordering) are positional errors a model can correct when told "your unitary does not equal the target"; F4 failures (recipe-verification / commitment) involve mis-tracking what the composition computes, and structured per-round feedback may be insufficient to redirect a model that has already committed to a wrong-recipe shape. The sample is small ($n = 5$ round-0 failures), so this is best read as *directional rather than confirmatory*: the observed pattern is consistent with the cognitive distinction between ordering errors and recipe mis-execution, but a larger sample is needed to confirm it. We frame this as a hypothesis worth testing at scale rather than a settled finding. Per-cell agent traces are released at `results/agent_runs/`.

5.4 Stochastic decoding is the right default for benchmarks of this kind

Several failures in our dataset only appear at non-zero temperature. Llama-3.3-70B passes the QFT-3 prompt at greedy ($T = 0$) but fails at all three stochastic seeds with the F2c subtype (spurious extra phases). Claude Haiku 4.5 fails 3 of 3 QFT-3 stochastic seeds despite passing greedy. The pattern argues against the common practice of benchmarking LLMs at greedy decoding alone: greedy is at best a single sample from the output distribution and systematically hides modes the model would produce under realistic sampling pressure. We pre-registered multi-seed evaluation in our protocol precisely to surface this kind of effect; the results support the choice.

6 Limitations

Sample size.

Twelve paired prompts is small for a comprehensive benchmark paper; we present this as a methodological contribution plus a case study, with the specific QFT-3 and

phase-flip-oracle findings as the headline results. A scaled-up paired-prompt corpus targeting roughly 30–50 prompts is a natural direction for future work.

Contamination and the OOD framing.

The prompts are original (not drawn from a public benchmark), which reduces verbatim-memorization risk. The underlying *algorithms*, however (QFT, Grover, Toffoli decomposition), are textbook and surely appear in pretraining, so our "distant out-of-distribution" framing refers to the gate-level Qiskit *expression* and the indexing conventions, not to the algorithms themselves. We tested whether failures reflect a specific prompt phrasing with a paraphrase-invariance check on QFT-3 (Section 4.4): 81% of failures persisted under a reworded prompt, so wording is not the main driver there. We did not repeat the check on every prompt, which remains future work.

Per-provider reproducibility.

Anthropic's Claude 4.x models deprecated the `temperature` parameter; the API uses adaptive sampling. We label our seeds for cache-key purposes but cannot guarantee bit-identical reruns at non-zero *nominal* temperature.

Free-tier rate limits.

Google AI Studio's free tier rate-limited stochastic Gemini calls. OpenRouter's free tier (50 requests / account / day) bounds the agent demo described in Section 5.3. Affected cells are reported as missing data rather than imputed. Replicating these cells on a paid tier would fill the gaps; the cache architecture means previously-completed cells are not re-billed on rerun.

Pairing validity and control difficulty (the central limitation).

Every cross-half comparison rests on the classical control stripping quantum-specific knowledge *while preserving comparable general-programming difficulty.* Our *structural* controls fail the second requirement: they pass 99.5% of the time, so passing one carries almost no information, and the 100% structural specificity over observed pairs (Table 2) is an upper bound rather than a measurement. We regard this as the paper's central limitation, and the difficulty-matched FFT control (Section 3.8) is our response to it — but we built that matched control for the QFT-3 case study *only.* For the other 11 prompts the specificity numbers should be read as upper bounds. Several prompts (the phase-flip oracle, the alternate Bell state) have difficulty that is inherently quantum — phase and relative-sign reasoning with no clean classical analog — so constructing matched controls there is genuinely harder and is the main item of future work. The per-prompt `pairing_justification` fields and the classical canonical solutions (NumPy primitives only, no quantum library or identity) are released so readers can audit each pairing.

The FFT control isolates ordering, not phase semantics.

QFT difficulty has two components: (1) ordering the H and controlled-phase gates, and (2) the relative-phase/interference semantics realized by those rotations. The classical

Cooley–Tukey FFT shares component (1) exactly, but it performs the same complex-twiddle arithmetic by rote and does not exercise Hilbert-space interference reasoning. So a model that passes the FFT control but fails the QFT has been isolated from *generic ordering difficulty*, but not cleanly separated into "gate-circuit ordering" versus "interference reasoning." The 50% figure should be read as "not explained by generic ordering difficulty," which is weaker than "purely interference-driven." This is precisely why the quantum-native class (Section 3.2), whose difficulty *is* the phase/interference component with no classical analog, is a necessary complement to the single matched control.

Taxonomy: automated, not human, inter-rater agreement.

Our inter-rater reliability (Section 3.9) is computed across the author's primary coding and independent *LLM* raters, not independent humans. The capable rater (GPT-5.5) reaches $\kappa = 0.68$, just below the pre-registered 0.7; a weaker model agrees less, and our planned cross-provider raters were lost to free-tier rate limits, so we could not triangulate across labs as intended. The κ should be read as evidence that the codebook is *substantially reproducible by a capable automated rater*, not as human-validated ground truth; a human re-coding is the natural next step. A share of failures also land in broad per-family buckets (e.g. F4_other) rather than the finely named subtypes, with the F2 (QFT-ordering) family the most thoroughly characterized.

7 Deviations from pre-registration

Our protocol, committed before data collection, commits to enumerating any deviation. The material ones:

- **Prompt source.** The protocol planned to draw quantum prompts from Qiskit HumanEval [1]. We instead authored 12 original paired prompts so each quantum task had a cleanly matched classical control — a departure from the protocol's "we do not author new quantum prompts."
- **Decoding.** We used three stochastic seeds, not the five planned. Anthropic's Claude 4.x also deprecated the `temperature` parameter, so the nominal T=0.7 Claude seeds draw from adaptive sampling rather than a controlled temperature.
- **Inter-rater coding.** The protocol planned two *human* coders. We instead use the author's primary coding plus independent *LLM* re-coders, blind to the primary (capable rater GPT-5.5: Cohen's $\kappa = 0.68$); human re-coding remains future work.
- **Pairing sign-off.** The separate `pairings.csv` sign-off artifact in the protocol was not produced; pairing rationale lives inline in each prompt's `pairing_justification` field.
- **Primary metric.** We retain the pre-registered specificity ratio but, on finding the structural controls near-ceiling, added a *post-hoc* difficulty-matched control for the QFT-3 case study and treat that as the informative analysis (Sections 3.8, 4.4). We flag this as exploratory rather than pre-registered.

8 Conclusion

When an LLM fails at quantum-code generation, whether the failure is "quantum-specific" is not a property of the failure alone — it depends on what you compare against. Across 9 models on a 12-prompt paired benchmark we observed 73 quantum-half failures. Against a structural control (matched on output shape), the control passes at all 71 failures where it was observed (100%) and they all look quantum-specific; but it passes 99.5% of cells overall, so this is an upper bound that reflects control easiness, not quantum-specificity. When we replace it, for our QFT-3 case study, with a *difficulty-matched* control — the classical FFT, which demands the same ordered composition of non-commuting operations with no quantum semantics — apparent specificity halves to 8 of 16 (50%) and splits by model: capable models fail the QFT while passing the classical ordering task (quantum-specific), whereas weaker models fail both (generic difficulty with ordered composition). The methodological takeaway — which we state as a caution, not a law — is that **a failure-attribution claim is only as strong as the difficulty of its control**, and a trivial control over-attributes failures to the domain under study. We also map *when* a difficulty-matched control can exist: for 44 of 63 failures the quantum target is classically trivial, so no such control exists and specificity is not measurable there. We do not claim those are quantum-specific — our own independent rater codes the alternate-Bell family as generic — only that the QFT-3 result (the one prompt we could control) gives 50% and that the rest await better controls.

The failures we observed form a qualitative catalog of recurring shapes — gate-cluster ordering, reversed iteration direction, spurious extra phases, asymmetric edge-case guarding, identity-circuit emission, and incoherent thinking, among others. Many share a broader pattern: *LLMs reproduce the visual surface form of a quantum-code recipe without verifying that the resulting unitary computes the claimed transformation.* A capable independent LLM re-coder, blind to the author's labels, agrees on these codings with substantial reliability (Cohen's $\kappa = 0.68$; Section 3.9).

Two of the nine models tested (Claude Sonnet 4.6 and GPT-5.5 — both current-generation frontier paid models) exhibit zero quantum-specific failures across the benchmark. The remaining seven models, including the code-specialized Qwen2.5-Coder-32B, exhibit between 2 and 24 failures each. The presence of failures even on a code-specialized model supports our interpretation that quantum-code generation requires quantum-specific cognitive skills beyond general code-generation competence; a code-LLM specialization is not sufficient.

A simulator-feedback agent applied to the failure cells provides preliminary evidence that the F2-family (gate-ordering) failures are more recoverable via structured grader feedback than the F4-family (recipe-verification / commitment) failures, consistent with the cognitive interpretation: gate-ordering is a positional error a model can fix when told "your gate sequence does not equal the target unitary," whereas recipe-verification involves mis-tracking what the composition computes, which a per-round error message is unlikely to correct.

Future work.

Three directions follow naturally from the present study: (1) scaling the paired-prompt corpus beyond the 12 prompts characterized here to roughly 30–50 prompts to support per-tier statistical claims; (2) extending the agent demo to the full panel $\times$ all rounds $\times$ multiple seeds, with per-failure-subtype recovery curves as the primary metric; and (3) generalizing the paired-prompt methodology beyond quantum to other distant out-of-distribution coding tasks (e.g., FPGA HDLs, blockchain smart contracts, formal verification) where the question of whether failures are domain-specific or generic is comparably underexplored.

Reproducibility statement

All paired prompts (`data/prompts/pilot/`, including the `classical_stress` half for QFT-3), model responses (`results/runs/`), per-cell taxonomy assignments (`results/taxonomy.json`), agent traces (`results/agent_runs/`), evaluation code (`src/eval.py`, `src/runner.py`, `src/stress.py`, `src/agent.py`), the statistics and modeling (`src/stats.py`, `src/modeling.py`, `src/report.py`), the ablations (`src/ablation.py`, `src/ablation_thinking.py`), the independent re-coding (`src/recode.py`), and the analysis pipeline (`src/analysis.py`) — with a 30-test suite (`tests/`) — are publicly available at https://github.com/mnn31/llm4quantum and archived on Zenodo at https://zenodo.org/records/20600905 (DOI: 10.5281/zenodo.20600905). Every cached response is keyed by $\text{SHA256}(\text{prompt}, \text{model}, T, \text{seed})$ and is exactly reproducible offline; the cache is idempotent across reruns and skips already-completed cells.

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Statements and Declarations

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Competing interests.

The author declares no competing financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

Ethics approval.

Not applicable. This study did not involve human participants, human data or tissue, or animals.

Clinical trial number.

Not applicable.

Data availability.

The paired-prompt dataset, all cached model responses, and the per-cell taxonomy codings that support the findings of this study are openly available in the LLM4QUANTUM repository at https://github.com/mnn31/llm4quantum, and archived on Zenodo at https://zenodo.org/records/20600905 (DOI: 10.5281/zenodo.20600905).

Code availability.

The complete evaluation, statistics, modeling, ablation, and analysis pipeline, together with the test suite, is openly available in the same repository; the Reproducibility statement above gives a file-level index.

Author contributions.

M.G. is the sole author and is responsible for all aspects of the work: conceptualization, methodology, software, formal analysis, verification, and writing, and approved the final manuscript.

Use of generative AI.

The use of generative AI tools in the research and writing process is documented in Section 3.11, in accordance with Springer Nature editorial policy.

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